The Role of Reason & Observation in Unveiling Natural Truths

Reason vs. Experience
Plato belittles senses while Aristotle integrates them

Celestial vs. Terrestrial
Aristotle differs them while Newton unifies them

Newton: Blends Plato's diductive rigor with Aristotle's empirical focus via math

Plato

- **World of Forms:** True Reality vs. Illusory Senses (Text 1a, 1b)
  - **Allegory of the Cave:** Knowledge as escape from sensory deception (Text 1a)
- **Deductive Reasoning:** Truth via philosophical reflection, not observation (Text 1b)
  - **Reminiscence:** Recollection of innate Ideas (Text 1b)

Aristotle

- **Metaphysics & Epistemology**
  - **Empiricism:** Knowledge begins with sensory experience (Text 2)
  - **Inductive Reasoning:** General truths from specific observations (Text 2)
- **Nature & Change**
  - Form & Matter (Text 2)
  - 4 causes
- **Cosmology & Motion**
  - **Natural Motion:** Objects move toward inherent purposes (Text 2)

Bridge
Medieval Kinematics (Text 2)

Newton

- **Unified Physics:** Celestrial/terrestrial laws merged (Text 3a)
  - Newtonian Dynamics (Text 3a)
- **Three Principia Steps:**
  - Definitions (reason)
  - Axioms (deduction)
  - Results (empirical validation) (Text 3b)